HDOC Day-45

Write a menu-driven C program to perform the following operations using separate functions.

- Change the date format from dd/mm/yyyy to dd-mmm-yy. **E.g.** Input: 12/10/2026, Output: 12-Oct-26
- Print all sub-strings of a string. **E.g.** Input: abc, Output: a,ab,abc,b,bc,c

1. Change date format from dd/mm/yyyy to dd-mmm-yy

Algorithm

1. Start.
2. Declare variables to store the day, month, and year.
3. Input the date in dd/mm/yyyy format.
4. Extract the day, month, and year from the input.
5. Convert the numeric month into its three-letter abbreviation using a separate function.
6. Convert the year into its last two digits.
7. Display the date in dd-mmm-yy format.
8. Stop.

Example: 12/10/2026 → 12-Oct-26

• C Program:

```
#include <stdio.h>
    #include <string.h>
// Function to change date format
    void changeDate()
    {
      int day, month, year;
      char date[20];
      char *months[] = {
        "Jan", "Feb", "Mar", "Apr", "May", "Jun",
        "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"
      };
```

```
    printf("Enter date (dd/mm/yyyy): ");
    scanf("%19s", date);

    if (sscanf(date, "%d/%d/%d", &day, &month, &year) != 3 ||
        strlen(date) != 10 ||
        date[2] != '/' || date[5] != '/' ||
        day < 1 || day > 31 ||
        month < 1 || month > 12 ||
        year < 1000 || year > 9999)
    {
        printf("Invalid date format!\n");
        return;
    }

    printf("Formatted date: %02d-%s-%02d\n",
           day, months[month - 1], year % 100);
}

// Function to print all substrings
void printSubstrings()
{
    char str[100];
    int i, j, k, len;

    printf("Enter a string (without spaces): ");
    scanf("%99s", str);

    len = strlen(str);

    printf("All substrings:\n");

    for (i = 0; i < len; i++)
    {
        for (j = i; j < len; j++)
        {
            for (k = i; k <= j; k++)
            {
                printf("%c", str[k]);
            }
            printf(", ");
```

```
        }
      }

      printf("\n");
    }

    int main()
    {
      int choice;

      do
      {
        printf("\n--- MENU ---\n");
        printf("1. Change date format\n");
        printf("2. Print all substrings\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
          case 1:
     changeDate();
            break;

          case 2:
            printSubstrings();
            break;

          case 3:
            printf("Program ended.\n");
            break;

          default:
            printf("Invalid choice!\n");
        }

      } while (choice != 3);

      return 0;
}
```

2. Print all substrings of a string

Algorithm

1. Start.
2. Input a string.
3. Find the length of the string.
4. Use an outer loop to select the starting position of a substring.
5. Use an inner loop to extend the substring from the starting position.
6. Print each substring.
7. Repeat until all possible substrings have been printed.
8. Stop.

- Example: Input: abc
- Output: a, ab, abc, b, bc, c

Test Case Table (1):

Test Case	Input Date	Expected Output
1	12/10/2026	12-Oct-26
2	25/12/2024	25-Dec-24

Test Case Table (2):

Test Case	Input String	Expected Output
1	abc	a, ab , abc , b , bc , c
2	cat	c , ca , cat , a , at , t
3	hello	hEx , he, hel , hell, hello, e ,el, ell , ello , l , ll , llo , l , lo , o

Compilation and Execution:

The following program has been compiled and executed successfully.

Test Case Table (1) Screenshots:

Test Case 1:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out

── MENU ──
1. Change date format
2. Print all substrings
3. Exit
Enter your choice: 1
Enter date (dd/mm/yyyy): 12/10/2026
Formatted date: 12-Oct-26
```

Test Case 2:

```
── MENU ──
1. Change date format
2. Print all substrings
3. Exit
Enter your choice: 1
Enter date (dd/mm/yyyy): 25/12/2024
Formatted date: 25-Dec-24
```

Test Table (2) Screenshots:

Test Case 1:

```
── MENU ──
1. Change date format
2. Print all substrings
3. Exit
Enter your choice: 2
Enter a string (without spaces): abc
All substrings:
a, ab, abc, b, bc, c,
```

Test Case 2:

```
Enter your choice: 2
Enter a string (without spaces): hello
All substrings:
h, he, hel, hell, hello, e, el, ell, ello, l, ll, llo, l, lo, o,
```

Test Case 3:

```
── MENU ──
1. Change date format
2. Print all substrings
3. Exit
Enter your choice: 2
Enter a string (without spaces): cat
All substrings:
c, ca, cat, a, at, t,
```